

# Greedy Positive-Cone Reduction: from Three Baselines to a Standard Angular-Defect Greedy

Experimental summary

July 24, 2026

## Abstract

We build a reduced *positive cone*  $K = \text{span}_+\{w_1, \dots, w_R\}$  approximating a set of non-negative snapshots, and compare three greedy constructions — Cone-Projected Greedy (CPG), Angular-Defect Greedy (ADG), modified CPG (MCPG) — on a low-dimensional *lambda* dataset and a high-dimensional *physics* contact dataset. The note is organised as the *evolution* of the method: a fair baseline comparison exposes two weaknesses of the raw greedy cones (poor worst-case relative accuracy when snapshot magnitudes vary, and severe ill-conditioning from near-duplicate generators); we then add one ingredient to ADG — *per-snapshot normalisation* — and call the result the **standard ADG**. Geometrically this ingredient is a single move (Sec. 2): it sends every snapshot to the unit sphere and turns the positive cone into a spherical region, so novelty becomes a geodesic distance and the whole method is read intrinsically, on directions alone. It addresses the first weakness and not the second. At equal cone size the standard ADG is 15–28 $\times$  more accurate than CPG/MCPG on the wide-magnitude physics data (magnitude ratio 604); on the well-scaled lambda data it selects the same generators as CPG, inherits the same  $\kappa_2 \sim 10^{10}$ , and MCPG dominates both. Conditioning is therefore MCPG’s contribution, not ADG’s. We then re-test the story on four independent contact-force datasets (membrane obstacle, and three Hertz-contact cases), in their dual  $W$ -inner product and with held-out parameters. They confirm the magnitude-ratio thesis on data it was not developed on, and force two refinements: normalisation is *harmful*, not merely inert, on well-scaled data — including on lambda, where it was first called inert; and CPG and ADG are *not* the same algorithm, despite selecting identically on every dataset the method was developed on.

## 1 Setup

**Algorithms.** All three methods grow a positive cone one generator at a time (ADG admits every snapshot tied at the maximal angle, so a round may add more than one) and project each snapshot  $x$  onto the current cone by non-negative least squares (NNLS),  $\Pi_K(x) = \arg \min_{c \geq 0} \|Ac - x\|_2$  with  $A$  the generators.

- CPG adds the snapshot of largest *absolute* residual  $\|x - \Pi_K(x)\|_2$ .
- ADG adds the snapshot of largest *angle*  $\theta_K(x) = \arcsin(\|x - \Pi_K(x)\|_2 / \|x\|_2)$  to the cone — a *scale-invariant* novelty measure, so its selection is intrinsically insensitive to snapshot magnitude. A round admits *every* snapshot tied at the maximal angle  $\theta_{\max}$ , and per-snapshot normalisation (Sec. 2) is its only remaining knob; the **standard ADG** is ADG with that knob on.
- MCPG selects like CPG but stores a shifted, re-normalised generator  $\nu = (x - \Pi_K(x)) / \|x - \Pi_K(x)\|_2$ , carrying only the direction new to the cone.

**Metric.** We report the *per-snapshot relative residual*  $\rho(x) = \|x - \Pi_K(x)\|_2 / \|x\|_2$ , its worst case  $\rho_{\max}$  and mean  $\bar{\rho}$  over the dataset, evaluated on the *original, physical-scale* snapshots for every method, plus the basis condition number  $\kappa_2 = \sigma_{\max} / \sigma_{\min}$ .

**Datasets.** Table 1 lists the six datasets. The magnitude ratio  $\max_x \|x\|_2 / \min_x \|x\|_2$  is the single quantity that predicts which method wins.

Table 1: Datasets. The magnitude ratio (over nonzero snapshots) governs the qualitative differences below. The four contact-force sets (Sec. 6) are measured in their dual  $W$ -norm. Every set here except physics is well-scaled (ratio  $\leq 4$ ), which by the thesis of this note predicts MCPG-like behaviour and a normalisation that costs more than it returns — borne out on lambda itself (Sec. 4) and on all four contact sets (Sec. 6). Hertz (a) is the only sample at an *intermediate* ratio, and Sec. 6.5 uses it to place a lower bound on where normalisation starts to pay.

| Dataset   | Snapshots $\times$ dim | $\min \ x\ $       | $\max \ x\ $       | Magnitude ratio |
|-----------|------------------------|--------------------|--------------------|-----------------|
| Lambda    | $50 \times 57$         | 0.283              | 0.319              | 1.13            |
| Physics   | $96 \times 7676$       | $9.04 \times 10^6$ | $5.46 \times 10^9$ | 604             |
| Hertz     | $81 \times 47$         | 1.036              | 1.204              | 1.16            |
| Membrane  | $125 \times 885$       | 1.043              | 1.353              | 1.30            |
| Hertz (a) | $31 \times 51$         | 0.123              | 0.482              | 3.93            |
| Hertz (b) | $23 \times 51$         | 0.101              | 0.109              | 1.09            |

## 2 The standard ADG on the unit sphere

The single ingredient that turns ADG into the *standard* ADG is per-snapshot normalisation. Algebraically it is small: divide each snapshot by its own norm before the stopping/unresolved check, while still storing *physical-scale* generators, so  $\varepsilon$  becomes a genuine per-snapshot relative tolerance rather than the absolute floor  $\varepsilon \max_x \|x\|$  shared across the dataset. It is applied to ADG only; CPG/MCPG have no such option and stay on the raw snapshots. It moves the stopping test, never the argmax: because the angle  $\theta_K$  is scale-invariant, normalisation cannot change *which* generator a round selects, only how many rounds run before every snapshot counts as resolved.

Geometrically that small move is one clean picture, and it is the whole of what “standard” means here (Figs. 1–2).

- **Snapshots live on the unit sphere.** Normalising  $x \mapsto \hat{x} = x / \|x\|$  sends every nonzero snapshot to a point of the unit sphere  $S^{d-1}$ : magnitude is discarded and only direction survives. The wide-magnitude cloud of the left panels collapses onto the sphere on the right.
- **The cone becomes a spherical region.** The positive cone  $K = \text{span}_+\{w_1, \dots, w_R\}$  is scale-invariant ( $x \in K \iff \hat{x} \in K$ ), so it is fully described by its trace  $\hat{A}(K) = K \cap S^{d-1}$  — an *arc* in  $\mathbb{R}^2$  (Fig. 1, right), a geodesically convex *spherical triangle/cap* in  $\mathbb{R}^3$  (Fig. 2, right). The generator directions  $\hat{w}_i$  are its vertices and  $\hat{A}(K)$  is their geodesic convex hull; its area (the solid angle read out in Fig. 2, 0.36 sr, 2.9% of the sphere) is the fraction of directions the cone can represent.

- **Novelty is a geodesic distance.** For a snapshot direction  $\hat{x}$ , the selection score  $\theta_K(x) = \arcsin(\|x - \Pi_K(x)\|_2 / \|x\|_2)$  is exactly the great-circle angle from  $\hat{x}$  to  $\hat{A}(K)$  (the residual  $x - \Pi_K(x)$  is orthogonal to  $\Pi_K(x)$ , so the triangle  $0, \Pi_K(x), x$  is right-angled). ADG therefore adds the snapshot whose direction is geodesically *farthest* from the current region — it patches the widest angular gap in the sphere’s coverage. Scale-invariance is manifest: two snapshots on one ray are the same point of  $S^{d-1}$  and are indistinguishable to the rule.
- **Standard ADG is the whole analysis done on the sphere.** Raw ADG scores its stopping residual against one global scale  $\varepsilon \max_x \|x\|$ , so a small-magnitude snapshot can drop below that absolute floor while its *direction* is still far from  $\hat{A}(K)$ . Normalisation deletes the magnitude axis: the stopping test becomes “is every snapshot direction within the target angular tolerance of  $\hat{A}(K)$ ?”, the intrinsic question the sphere makes obvious. On wide-magnitude data (physics, ratio 604) this rescues the small snapshots the floor hides; on well-scaled data (ratio  $\approx 1$ ) every point already sits in a thin band on the sphere, so normalisation changes only the bookkeeping (Sec. 4).
- **The certificate is an angular defect.** The worst residual after a round is bounded by  $\max(\varepsilon C, C \sin \theta_{\max})$  with  $C = \max_x \|x\|$  and  $\theta_{\max}$  the largest remaining geodesic gap — the *angular defect* that names the method. The bound is checked against the recorded convergence history at every round (`verify_angular_defect_certificate`).

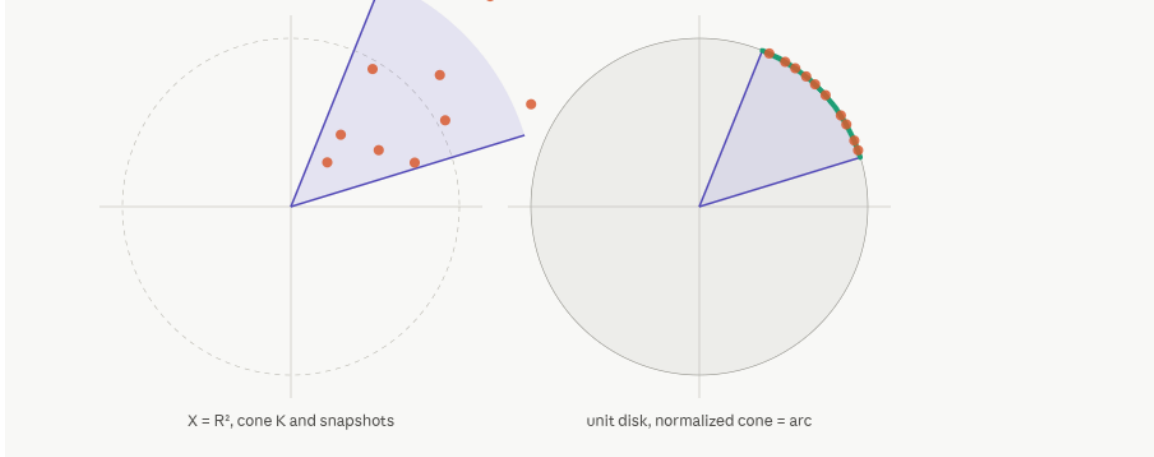


Figure 1:  $\mathbb{R}^2$ : a positive cone  $K$  (purple wedge) and snapshots (orange). Normalising sends each snapshot to the unit circle and  $K$  to the *arc*  $\hat{A}(K) = K \cap S^1$  (right). The stopping test asks that every snapshot direction lie within an angular tolerance of that arc; the selection score is the great-circle gap to it.

### 3 Baseline: the three methods as-is

Run at a common relative tolerance  $\varepsilon = 10^{-2}$  (each method stops at its own cone size), the raw greedy cones already reveal the two problems that motivate everything else (Table 2). **This table is diagnostic, not a fair comparison:** a common  $\varepsilon$  stops the three methods at *different*  $R$  (lambda 23/15/23, physics 9/9/11), so any accuracy gap here conflates the method with how many generators its tolerance happened to buy. The fair, like-for-like comparison — every method at the

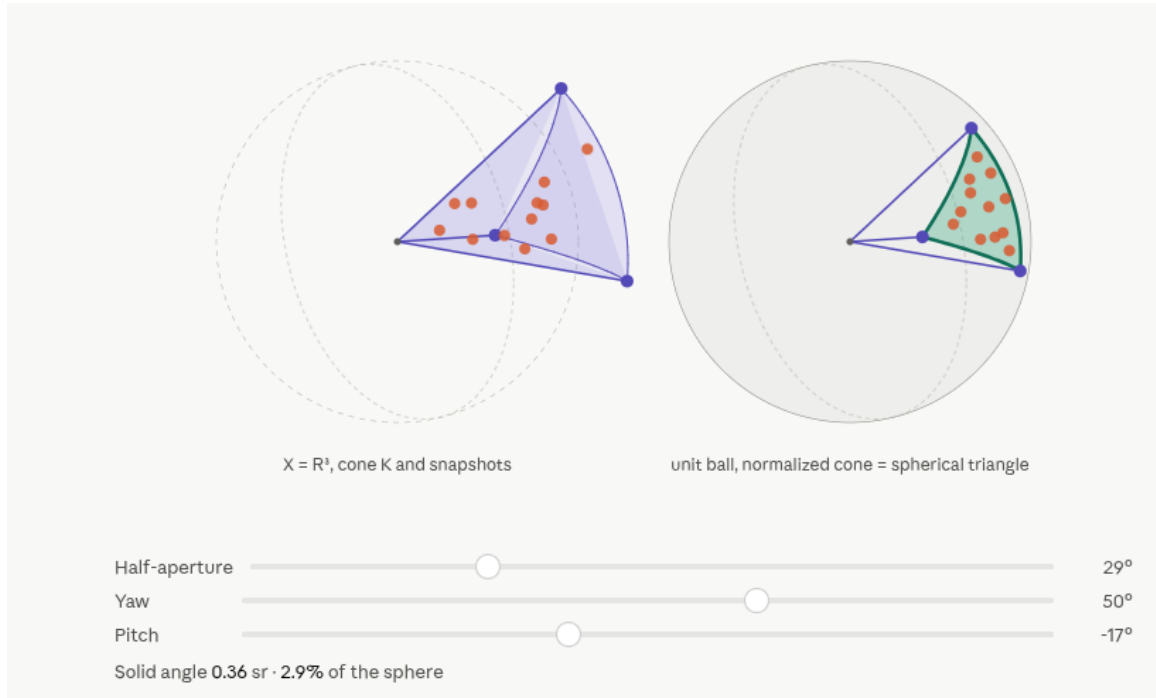


Figure 2:  $\mathbb{R}^3$ : the same picture one dimension up. The cone's trace on the unit sphere is a geodesically convex *spherical triangle*  $\hat{A}(K)$  (right, green); its solid angle (0.36 sr here, 2.9% of the sphere) measures how much of direction-space the cone spans. ADG grows  $\hat{A}(K)$  by admitting the snapshot direction geodesically farthest from it.

same cone size  $R$  — is Table 3; read the numbers below only as evidence of the two failure modes, never as “method A beats method B”.

Table 2: Baseline at  $\varepsilon = 10^{-2}$  (whole dataset). Physics: ADG’s scale-invariant selection halves the worst-case error of CPG/MCPG. Lambda: MCPG is best and, crucially, CPG/ADG are numerically singular ( $\kappa_2 \sim 10^{10}$ ).

| Dataset | Method | $R$ | $\rho_{\max}$         | $\kappa_2$            |
|---------|--------|-----|-----------------------|-----------------------|
| Lambda  | CPG    | 23  | $1.10 \times 10^{-2}$ | $5.43 \times 10^{10}$ |
|         | MCPG   | 15  | $8.90 \times 10^{-3}$ | $1.10 \times 10^3$    |
|         | ADG    | 23  | $1.10 \times 10^{-2}$ | $5.43 \times 10^{10}$ |
| Physics | CPG    | 9   | $8.14 \times 10^{-1}$ | $2.10 \times 10^2$    |
|         | MCPG   | 9   | $8.06 \times 10^{-1}$ | $2.18 \times 10^2$    |
|         | ADG    | 11  | $3.53 \times 10^{-1}$ | $1.39 \times 10^3$    |

**Two weaknesses.** (i) **Accuracy on wide-magnitude data.** On physics, CPG/MCPG rank candidates by *absolute* residual, so they chase a few high-magnitude snapshots and leave the small-magnitude ones badly resolved ( $\rho_{\max} \approx 0.8$ ); ADG’s angle criterion already does better (0.35), but not enough. (ii) **Conditioning.** On lambda the snapshots are nearly collinear, so raw-snapshot cones (CPG, ADG) accumulate near-parallel generators and become singular ( $\kappa_2 \approx 5 \times 10^{10}$ ). Only the first of these is an ADG problem with an ADG fix, and Sec. 4 gives it. The second is structural: any method that stores raw snapshots as generators inherits it, so it is answered by MCPG’s shifted, re-normalised generators rather than by anything ADG can do to its selection rule.

## 4 What normalisation buys, and what it costs

Sec. 2 defined the operation and its geometry; this section is what it does to the numbers, on the two datasets it was developed on. The rule of thumb is the magnitude ratio: normalisation rescues masked small-magnitude snapshots when the ratio is large, and merely demands more generators when it is near 1.

Figure 3 shows the effect on physics: the raw methods leave a broad plateau of small-magnitude snapshots stuck near  $10^{-2}$  relative error, while normalised ADG drives them to machine precision. At  $\varepsilon = 10^{-2}$  normalised ADG reaches  $\rho_{\max} = 0.143$  (mean  $3.7 \times 10^{-3}$ ) versus 0.35 for raw ADG and 0.81 for CPG/MCPG.

On lambda (magnitude ratio 1.13) it is *not* free, and calling it “inert” there — as an earlier version of this note did — understates the cost: it inflates  $R$  from 23 to 26 and  $\kappa_2$  from  $5.43 \times 10^{10}$  to  $1.38 \times 10^{11}$ , buying a  $\rho_{\max}$  of  $9.52 \times 10^{-3}$  against  $1.10 \times 10^{-2}$  — a 13% accuracy gain for 13% more generators and 2.5× worse conditioning. That is the same pattern the contact-force sets show (Sec. 6), so it is not a quirk of those datasets: whenever the magnitude ratio is near 1 there is no masked small-magnitude snapshot to rescue, and the stricter criterion only demands more generators. Normalisation earns its keep on physics (ratio 604), not here.

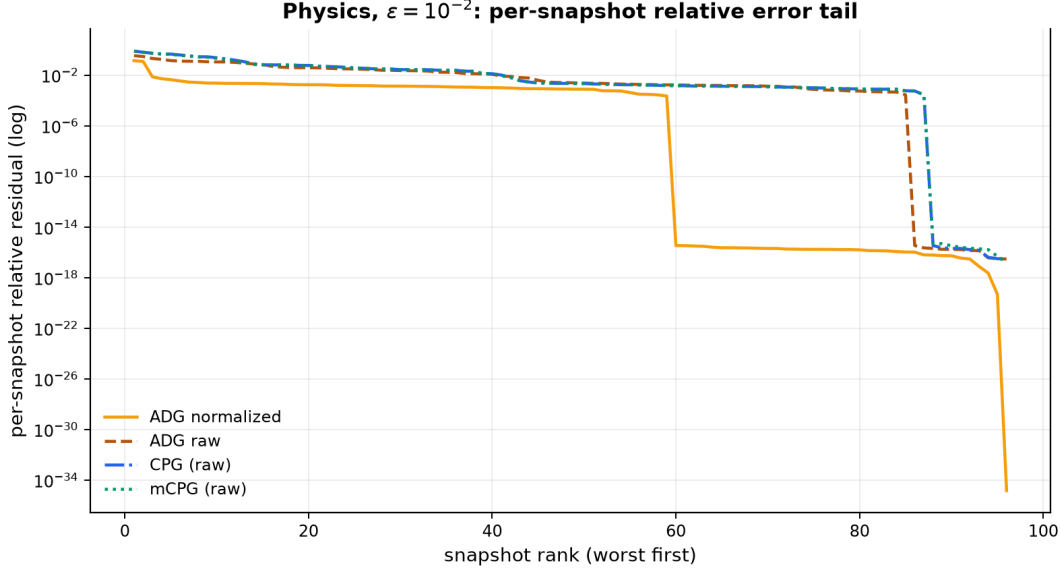


Figure 3: Physics,  $\varepsilon = 10^{-2}$ : per-snapshot relative residual sorted worst-first (log scale). Normalisation (applied to ADG only) resolves the small-magnitude snapshots that a global-scale criterion neglects, which is what collapses  $\rho_{\max}$ .

## 5 Fair comparison at matched cone size

**This is the fair comparison, and the one to read.** A method is only worth comparing to another at the *same budget*: equal cone size  $R$ , i.e. equal number of generators. Because normalisation moves ADG’s stopping test but never its  $\arg\max$  (Sec. 2), a common  $\varepsilon$  stops the three methods at different  $R$ , so their own stopping points measure the tolerance as much as the method (Table 2). We remove that confound by walking each basis prefix  $K_1, \dots, K_R$  and comparing at *matched*  $R$  (Table 3); all three greedies are nested, so a prefix of length  $n$  is exactly the cone that method would have produced had  $\varepsilon$  stopped it at  $n$ . Every conclusion in this note rests on these equal- $R$  numbers, not on the equal- $\varepsilon$  baseline. The rows below were regenerated from the current pipelines and match to all reported digits.

**Reading the comparison.** On **physics** the standard ADG dominates on accuracy at every budget: its worst-case relative error is 15–28 $\times$  lower than CPG/MCPG ( $1.87 \times 10^{-2}$  versus  $5.20 \times 10^{-1}$  at  $R = 30$ ), because only ADG, through normalisation, attends to the small-magnitude snapshots the other two leave 50% wrong. Note what it does *not* buy: ADG’s cone is no better conditioned than CPG’s, and at small  $R$  it is an order of magnitude *worse* ( $1.3 \times 10^3$  against  $1.0 \times 10^2$  at  $R = 5$ ). Accuracy on wide-magnitude data is the whole of the claim.

On **lambda** the standard ADG and CPG are indistinguishable — same  $\rho_{\max}$  at every  $R$ , same  $\kappa_2$  to within rounding — because with a magnitude ratio of 1.13 there is no masked snapshot for the angle criterion to rescue, and the two selection rules agree. Both become numerically singular past  $R \approx 21$ . MCPG is the only method that is both more accurate ( $6.55 \times 10^{-3}$  at  $R = 30$ , an order of magnitude better) and better conditioned. On well-scaled data ADG has nothing to add over CPG, and the right choice is MCPG.

Table 3: Matched cone size: worst-case relative residual  $\rho_{\max}$  and condition number  $\kappa_2$  for the standard ADG and for CPG/MCPG at the same  $R$  (whole dataset). Physics is the wide-magnitude set (ratio 604) and lambda the well-scaled one (ratio 1.13); the two behave in opposite ways. Source: `physics_component_sweep_metrics.csv`, `component_sweep_metrics.csv`.

| Dataset | $R$ | ADG                   |                       | CPG                   |                       | MCPG                  |                       |
|---------|-----|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|         |     | $\rho_{\max}$         | $\kappa_2$            | $\rho_{\max}$         | $\kappa_2$            | $\rho_{\max}$         | $\kappa_2$            |
| Physics | 5   | $1.66 \times 10^{-1}$ | $1.30 \times 10^3$    | $8.43 \times 10^{-1}$ | $1.00 \times 10^2$    | $8.43 \times 10^{-1}$ | $9.60 \times 10^1$    |
|         | 12  | $5.06 \times 10^{-2}$ | $1.60 \times 10^3$    | $7.63 \times 10^{-1}$ | $3.30 \times 10^2$    | $7.63 \times 10^{-1}$ | $2.50 \times 10^2$    |
|         | 24  | $2.43 \times 10^{-2}$ | $2.20 \times 10^3$    | $5.20 \times 10^{-1}$ | $1.20 \times 10^3$    | $5.20 \times 10^{-1}$ | $3.90 \times 10^2$    |
|         | 30  | $1.87 \times 10^{-2}$ | $2.30 \times 10^3$    | $5.20 \times 10^{-1}$ | $2.50 \times 10^3$    | $5.20 \times 10^{-1}$ | $3.70 \times 10^3$    |
| Lambda  | 7   | $7.98 \times 10^{-2}$ | $1.50 \times 10^2$    | $7.98 \times 10^{-2}$ | $1.50 \times 10^2$    | $6.60 \times 10^{-2}$ | $1.60 \times 10^1$    |
|         | 15  | $6.75 \times 10^{-2}$ | $4.00 \times 10^4$    | $6.75 \times 10^{-2}$ | $4.00 \times 10^4$    | $4.91 \times 10^{-2}$ | $4.60 \times 10^2$    |
|         | 21  | $6.75 \times 10^{-2}$ | $4.60 \times 10^{10}$ | $6.75 \times 10^{-2}$ | $4.00 \times 10^{10}$ | $9.62 \times 10^{-3}$ | $3.00 \times 10^{10}$ |
|         | 30  | $6.75 \times 10^{-2}$ | $4.80 \times 10^{11}$ | $6.75 \times 10^{-2}$ | $4.80 \times 10^{11}$ | $6.55 \times 10^{-3}$ | $7.10 \times 10^{10}$ |

## 6 Confirmation on four contact-force datasets

The three ideas above were developed on *lambda* and *physics*. We now test them on four independent datasets — the membrane-obstacle and Hertz-contact test cases of Niakh, Drouet, Ehrlacher & Ern (ESAIM:M2AN 2022), plus two further reconstructions of the Hertz physics, cases (a) and (b) — where  $\lambda(\mu) \geq 0$  is the physical contact force. Sections 6 –6.4 report the first two, on which the analysis was done; Sec. 6.5 adds cases (a) and (b). Together they confirm the note’s central thesis and force *three* refinements — the third, from case (a), retracting a claim the other three datasets had made look like a law.

**What is different here.** Three things depart from the lambda and physics analysis above, and each matters for reading the numbers.

- **A non-Euclidean dual inner product.** Errors, projections and angles are measured in  $\|x\|_W = \sqrt{x^\top W x}$  with  $W$  the dual mass matrix shipped with each dataset, not in  $\|\cdot\|_2$ . We factor  $W = U^\top U$  and run the *unmodified* Euclidean algorithms on  $U$ -transformed snapshots:  $\|x\|_W = \|Ux\|_2$  and  $U(Ac) = (UA)c$  with the same  $c \geq 0$ , so the transformed runs *are* the  $W$ -norm greedies. The lone exception is MCPG’s cone-shift constraint  $\Upsilon \leq \theta$ , which is elementwise and *not* preserved by  $U$ ; MCPG is given  $U^{-1}$  so that bound is imposed in physical coordinates.
- **Held-out testing.** Unlike Tables 2 and 3, which score the whole dataset, these runs fit on a training set and report a held-out  $\rho_{\max}$  at unseen  $\mu$ . Hertz holds out interior  $R_2$  values (65/16); the membrane holds out the strict interior of its  $5 \times 5 \times 5$  grid (98/27), since its  $\mu_1$  takes only 5 distinct values and a split on it would be degenerate. This turns out to change the conclusion (Sec. 6.4).
- **Feasible tolerances.**  $\varepsilon = 10^{-2}$  is unreachable on the membrane: POD needs  $R = 97/98$  to resolve the training set at all, so every method simply exhausts it ( $R = P$ ). The membrane is therefore run at  $\varepsilon = 0.25$ . Hertz uses  $\varepsilon = 10^{-2}$  as before.

### 6.1 Baseline: the lambda pattern, reproduced exactly

Table 4: Baseline on the contact-force sets (ADG raw; Hertz  $\varepsilon = 10^{-2}$ , membrane  $\varepsilon = 0.25$ ;  $\rho$  held out at unseen  $\mu$ ). Hertz reproduces the lambda failure mode exactly: CPG/ADG are numerically singular ( $\kappa_2 \sim 10^9$ ) while MCPG is compact and well-conditioned.

| Dataset  | Method | $R$ | $\rho_{\max}$ (train) | $\rho_{\max}$ (test)  | $\kappa_2$         |
|----------|--------|-----|-----------------------|-----------------------|--------------------|
| Hertz    | CPG    | 38  | $1.11 \times 10^{-2}$ | $6.28 \times 10^{-2}$ | $5.51 \times 10^9$ |
|          | MCPG   | 25  | $1.13 \times 10^{-2}$ | $4.82 \times 10^{-2}$ | $9.79 \times 10^2$ |
|          | ADG    | 38  | $1.11 \times 10^{-2}$ | $6.28 \times 10^{-2}$ | $5.51 \times 10^9$ |
| Membrane | CPG    | 56  | $2.98 \times 10^{-1}$ | $3.56 \times 10^{-1}$ | $4.91 \times 10^1$ |
|          | MCPG   | 52  | $2.71 \times 10^{-1}$ | $3.67 \times 10^{-1}$ | $3.07 \times 10^1$ |
|          | ADG    | 56  | $2.98 \times 10^{-1}$ | $3.56 \times 10^{-1}$ | $4.91 \times 10^1$ |

Table 4 is Table 2’s lambda block again, on new data: CPG and ADG are *bit-identical* (they select the same set, in a different order, hence the same cone) and near-singular at  $\kappa_2 = 5.5 \times 10^9$ ; MCPG reaches the same tolerance with 34% fewer generators and a  $\kappa_2$  seven orders of magnitude smaller. Both datasets have magnitude ratio  $\approx 1.2$ , and both behave exactly as the ratio predicts. This is the note’s main claim confirmed on data it was not developed on. As in Table 2, these are each method’s own  $\varepsilon$  stopping points ( $R = 38/25/38$  on Hertz), so the table is again diagnostic only; the fair equal- $R$  comparison is Table 5.

### 6.2 Normalisation on new data: not merely inert — harmful

Sec. 4 found normalisation “inert” on uniform-magnitude data. On these sets it is worse than inert: at ratios of 1.16/1.30 there is no masked small-magnitude snapshot for it to rescue, so the stricter per-snapshot criterion only demands more generators for no return. On Hertz it takes ADG from  $R = 38$  to  $R = 42$  and  $\kappa_2$  from  $5.5 \times 10^9$  to  $4.6 \times 10^{11}$ ; on the membrane from  $R = 56$  to  $R = 82$ . Held-out error is unchanged to three digits in both (`adg_variants.csv`). Normalisation here costs 11% more generators and two orders of magnitude of conditioning, and returns nothing.

*Refinement.* Normalisation should be read as conditional on the magnitude ratio, not as free: switch it on for wide-magnitude data (physics, ratio 604), off otherwise. Since the standard ADG is normalisation, this says the standard ADG is itself conditional — it is the right method only in the wide-magnitude regime.

### 6.3 Matched cone size

Because all three greedies are nested, the prefix  $K_1 \subset \dots \subset K_R$  is exactly the cone each would return at a looser  $\varepsilon$ . Walking the prefixes therefore separates “better cone” from “stopped sooner” (Table 5, Fig. 4).

At matched  $R$  the standard ADG buys *nothing* here: it is CPG, to every digit in the table, because at a magnitude ratio of 1.16 its normalisation has no masked snapshot to rescue and its angle criterion picks CPG’s set. The membrane agrees (at  $R = 52$ : MCPG  $\kappa_2 = 3.07 \times 10^1$ ,  $\rho_{\max}^{\text{test}} = 3.673 \times 10^{-1}$ ; ADG  $4.25 \times 10^1$ ,  $3.686 \times 10^{-1}$ ; CPG worst on test at  $3.786 \times 10^{-1}$ ). On well-scaled data MCPG is the method of choice at every budget; ADG has no role.

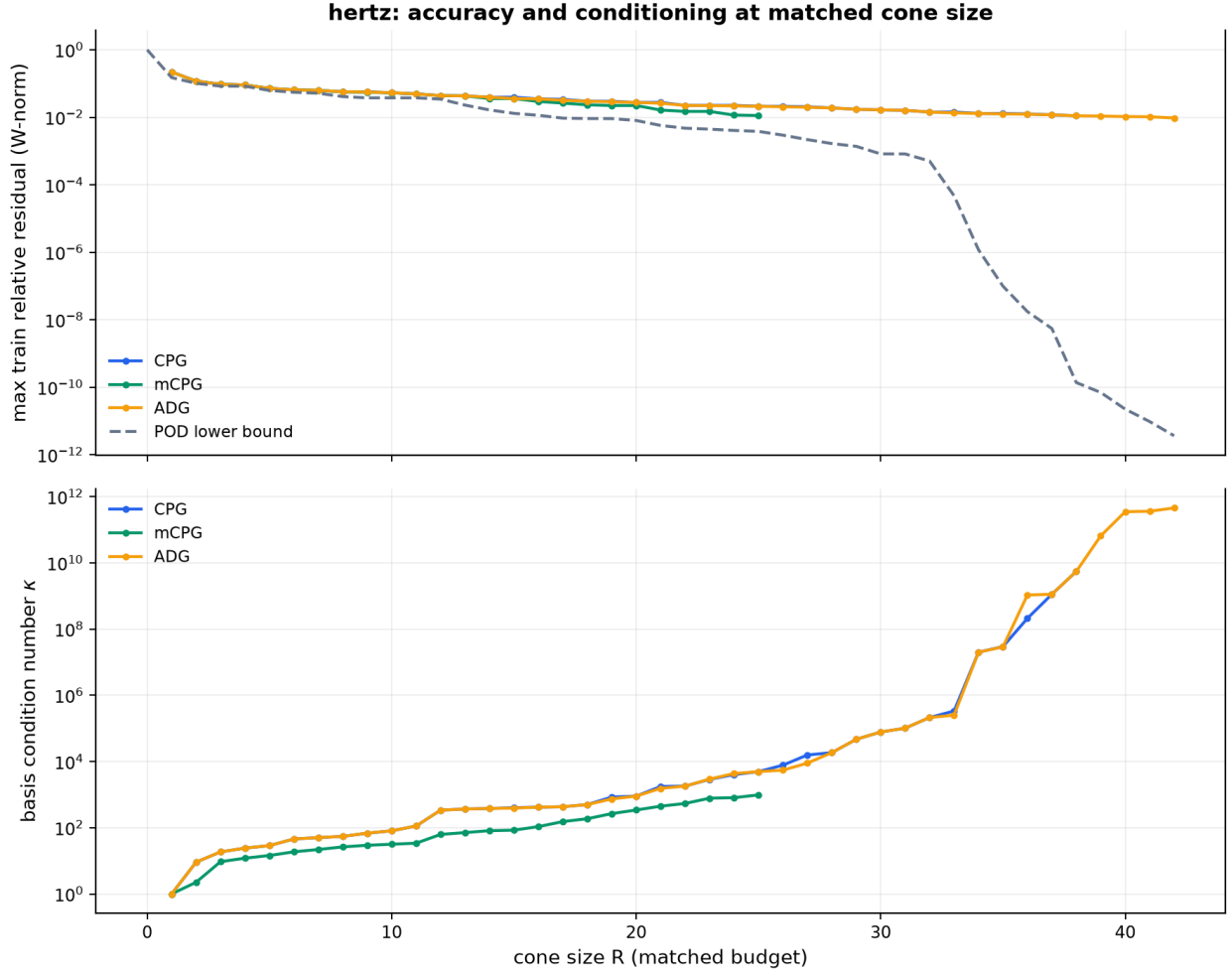


Figure 4: Hertz: accuracy (top) and conditioning (bottom) at matched cone size. mCPG (green) holds a  $\sim 4\times$  conditioning margin at every  $R$  and stays near  $10^3$ , while CPG/ADG (blue/orange, exactly overlapping) blow up past  $R \approx 32$  to  $10^{10}$ .

Table 5: Matched cone size on Hertz ( $\varepsilon = 10^{-2}$ ). At equal  $R$ , MCPG dominates on conditioning and on both error columns. CPG and ADG are identical at every  $R$  *on this dataset* — they select the same set in a different order — but that coincidence is not general (Sec. 6.5). Source: `matched_r.csv`.

| $R$ | Method | $\rho_{\max}$ (train) | $\rho_{\max}$ (test)  | $\kappa_2$         |
|-----|--------|-----------------------|-----------------------|--------------------|
| 5   | CPG    | $7.17 \times 10^{-2}$ | $6.94 \times 10^{-2}$ | $2.88 \times 10^1$ |
|     | MCPG   | $7.17 \times 10^{-2}$ | $6.94 \times 10^{-2}$ | $1.45 \times 10^1$ |
|     | ADG    | $7.17 \times 10^{-2}$ | $6.94 \times 10^{-2}$ | $2.88 \times 10^1$ |
| 12  | CPG    | $4.38 \times 10^{-2}$ | $6.28 \times 10^{-2}$ | $3.40 \times 10^2$ |
|     | MCPG   | $4.38 \times 10^{-2}$ | $6.10 \times 10^{-2}$ | $6.27 \times 10^1$ |
|     | ADG    | $4.38 \times 10^{-2}$ | $6.28 \times 10^{-2}$ | $3.40 \times 10^2$ |
| 16  | CPG    | $3.50 \times 10^{-2}$ | $6.28 \times 10^{-2}$ | $4.17 \times 10^2$ |
|     | MCPG   | $2.94 \times 10^{-2}$ | $6.04 \times 10^{-2}$ | $1.08 \times 10^2$ |
|     | ADG    | $3.50 \times 10^{-2}$ | $6.28 \times 10^{-2}$ | $4.17 \times 10^2$ |
| 25  | CPG    | $2.11 \times 10^{-2}$ | $6.28 \times 10^{-2}$ | $4.94 \times 10^3$ |
|     | MCPG   | $1.13 \times 10^{-2}$ | $4.82 \times 10^{-2}$ | $9.79 \times 10^2$ |
|     | ADG    | $2.11 \times 10^{-2}$ | $6.28 \times 10^{-2}$ | $4.94 \times 10^3$ |

#### 6.4 What only held-out testing reveals: a generalisation floor

Every lambda and physics table above scores the training data (there, the whole dataset). On Hertz that would have hidden the most important effect on these datasets: **the held-out error floors**. CPG/ADG’s test  $\rho_{\max}$  is pinned at  $6.280 \times 10^{-2}$  from  $R = 12$  all the way to  $R = 38$  — the last 26 generators improve training error  $4\times$  and generalisation by *nothing*. MCPG is the only method that keeps improving out of sample ( $6.10 \times 10^{-2}$  at  $R=12$ ,  $6.04 \times 10^{-2}$  at  $R=16$ ,  $4.82 \times 10^{-2}$  at  $R=25$ ).

The floor is the practical result of this section. It says the useful budget on Hertz is  $R \approx 12$ , and that everything past it is training-set decoration:  $\varepsilon$  is being asked for an accuracy the data cannot generalise to. It also explains why a smaller cone can look free on this dataset — past the floor, generators cost nothing observable to discard, so any procedure that stops early appears to lose nothing while in fact converging to a training error it never reaches out of sample.

**Why the membrane is hard.** Even a cone built from *all* 98 training snapshots leaves a held-out  $\rho_{\max} \approx 0.355$ , against  $\approx 0.12$  for an unconstrained POD subspace of the same data. Nonnegative combinations can only add mass, never cancel it, so a reaction bump at an unseen  $(\mu_2, \mu_3)$  is unreachable from bumps centred elsewhere. This is the moving-support regime, and no knob studied here addresses it.

#### 6.5 Two further Hertz cases: where CPG $\equiv$ ADG breaks

Two additional reconstructions of the Hertz physics sharpen the picture. Both put  $\lambda$  on the same reference arc (51 uniform nodes on  $[-1, 1]$ ) but drive it differently: **case (a)** sweeps the *imposed displacement*  $\mu = d \in [0.15, 0.45]$  (31 snapshots, held out 25/6), **case (b)** the *geometry*  $\mu = R_2 \in [0.905, 1.125]$  (23 snapshots, 18/5). Pressing harder in (a) both scales the semi-ellipse up

and widens its support, which is why (a) is the only set here with a non-trivial magnitude ratio (3.93, Table 1); (b)’s narrow range about  $R_2 \approx 1$  barely moves the profile (ratio 1.09).

**A caveat on the metric.** Unlike the other two sets, neither ships a  $W$  matrix. Both live on a 1-D contact arc, so  $W$  is rebuilt as the  $P_1$  arc mass matrix of the node abscissas — the same construction that produced the Hertz set’s own  $W$ . Without it the “ $W$ -norm” would silently degrade to a mesh-dependent nodal norm, so every number below rests on that reconstruction.

**They do not discriminate on accuracy.** Both are far lower-rank than Hertz (numerical rank 14 and 8; POD reaches  $1.8 \times 10^{-6}$  by  $R = 4$  on (b)). At  $\varepsilon = 10^{-2}$  all three methods hit the tolerance at  $R = 7$  (a) /  $R = 3$  (b) with *identical* errors, so Table 6 reports the discriminating  $\varepsilon = 10^{-3}$ . Conditioning is the only axis that separates the methods — and MCPG wins it by one to two orders, as on Hertz. These two sets are best read as a conditioning benchmark, not as evidence about accuracy.

Table 6: Hertz cases (a) and (b) at  $\varepsilon = 10^{-3}$  ( $\rho$  held out at unseen  $\mu$ ). Accuracy ties across methods; only  $\kappa_2$  separates them, and MCPG wins it everywhere. Case (b)’s MCPG cone is very nearly orthogonal.

| Dataset   | Method | $R$ | $\rho_{\max}$ (train) | $\rho_{\max}$ (test)  | $\kappa_2$         |
|-----------|--------|-----|-----------------------|-----------------------|--------------------|
| Hertz (a) | CPG    | 8   | $1.25 \times 10^{-3}$ | $1.44 \times 10^{-2}$ | $1.84 \times 10^3$ |
|           | MCPG   | 8   | $9.49 \times 10^{-4}$ | $1.44 \times 10^{-2}$ | $3.71 \times 10^1$ |
|           | ADG    | 8   | $1.25 \times 10^{-3}$ | $1.44 \times 10^{-2}$ | $1.84 \times 10^3$ |
| Hertz (b) | CPG    | 3   | $3.06 \times 10^{-4}$ | $2.97 \times 10^{-4}$ | $1.78 \times 10^2$ |
|           | MCPG   | 3   | $3.06 \times 10^{-4}$ | $2.97 \times 10^{-4}$ | 2.40               |
|           | ADG    | 3   | $3.06 \times 10^{-4}$ | $2.97 \times 10^{-4}$ | $1.78 \times 10^2$ |

**Refinement 3: CPG and ADG are not the same algorithm.** Table 4 and Table 5 record CPG and ADG as bit-identical, and lambda said the same; it is tempting to read that as a law. It is not. On case (a) at  $\varepsilon = 10^{-3}$  they again coincide — both select  $\{0, 3, 6, 10, 15, 20, 25, 30\}$ , differing only in order — but at  $\varepsilon = 0.05$  they *diverge*, and ADG *loses*:

| Method | Selections ( $R = 4$ ) | $\rho_{\max}$ (train) | $\rho_{\max}$ (test)  | $\kappa_2$         |
|--------|------------------------|-----------------------|-----------------------|--------------------|
| CPG    | $\{0, 10, 20, 30\}$    | $7.36 \times 10^{-2}$ | $3.93 \times 10^{-2}$ | $3.16 \times 10^1$ |
| MCPG   | $\{0, 10, 20, 30\}$    | $7.36 \times 10^{-2}$ | $3.89 \times 10^{-2}$ | $1.41 \times 10^1$ |
| ADG    | $\{0, 15, 20, 30\}$    | $1.03 \times 10^{-1}$ | $8.26 \times 10^{-2}$ | $3.76 \times 10^1$ |

The angular criterion prefers node 15 over node 10 and that is the wrong call:  $2.1\times$  worse held-out error at the same budget, and worse conditioning too. So the identity is a property of these particular datasets, not of the two algorithms — it holds while  $\varepsilon$  is tight enough that both are forced onto essentially the whole training set, and breaks as soon as  $\varepsilon$  is loose enough for the choice to bite. Where they differ, plain ADG is the one that loses. Across all six datasets and every  $\varepsilon$  tested, plain ADG never once beats CPG: it ties or it loses.

**Normalisation: a lower bound on where it pays.** Case (a)’s ratio of 3.93 is the first intermediate sample between the well-scaled sets (1.1–1.3) and physics (604). Normalisation is still *harmful* there:  $R$  goes  $8 \rightarrow 9$  and  $\kappa_2$   $1.8 \times 10^3 \rightarrow 1.6 \times 10^5$  at unchanged held-out error. A ratio near 4 is therefore not enough to make the per-snapshot criterion pay; the threshold lies well above it, closer to physics’ regime than to these.

**The floor reappears.** Case (a) reproduces Sec. 6.4’s effect: held-out  $\rho_{\max}$  is pinned at  $1.438 \times 10^{-2}$  from  $R = 7$  onward — for CPG and MCPG alike, and for every ADG variant that runs as far as  $R = 11$  — while training error keeps falling. That the floor is common to all methods and variants on a second, independently generated dataset supports reading it as a property of the interpolation split rather than of any one greedy.

## 7 Conclusions

- **Geometry.** The standard ADG is ADG read on the unit sphere (Sec. 2): normalisation sends snapshots to  $S^{d-1}$  and the cone to a spherical region  $\hat{A}(K)$ , making novelty a geodesic distance and the stopping test an intrinsic “every direction covered to tolerance” check. This is exactly why it helps when magnitudes vary and is inert when they do not.
- **Baseline.** Raw greedy cones have two failure modes: poor worst-case relative accuracy when snapshot magnitudes vary (CPG/MCPG on physics), and near-singular bases from near-parallel generators (CPG/ADG on lambda).
- **Normalisation** fixes the first for ADG, and defines the **standard ADG**: on physics it cuts the worst-case relative error from 0.52 (CPG/MCPG) to 0.019 at a matched  $R = 30$ , a  $28\times$  margin that holds at every budget.
- **Nothing in ADG fixes the second.** The conditioning failure is structural — any cone that stores raw snapshots as generators inherits it — so it is MCPG’s shifted, re-normalised generators that answer it, not ADG. On lambda the standard ADG is numerically singular at large  $R$  exactly like CPG ( $\kappa_2 \approx 5 \times 10^{11}$  at  $R = 30$ ), while MCPG is both an order of magnitude more accurate and better conditioned.
- **Standard ADG** is therefore a specialist, not a default: best on wide-magnitude data at every matched cone size, and indistinguishable from CPG everywhere else. *Guidance:* use the standard ADG when snapshot magnitudes span orders of magnitude; use MCPG otherwise — on well-scaled data it leads on both accuracy and conditioning, and ADG adds nothing over CPG.
- **Confirmed, with two refinements** (Sec. 6). Four independent contact-force datasets reproduce the lambda pattern exactly — magnitude ratio  $\approx 1.2$ , CPG/ADG identical and singular ( $\kappa_2 = 5.5 \times 10^9$ ), MCPG compact and well-conditioned — so the magnitude ratio predicts the winner on data the method was not tuned on. But: (i) normalisation is *harmful* rather than inert when the ratio is near 1 ( $R$ :  $38 \rightarrow 42$ ,  $\kappa_2$ :  $5.5 \times 10^9 \rightarrow 4.6 \times 10^{11}$  on Hertz), so the standard ADG is itself conditional on the regime; (ii) **CPG  $\equiv$  ADG is not a law** (Sec. 6.5). It holds on lambda, Hertz and Hertz (b), but breaks on Hertz (a) at  $\varepsilon = 0.05$ , where ADG makes a different — and  $2.1\times$  worse — held-out choice at the same  $R = 4$ . Plain ADG ties or loses to CPG on every dataset tested; it never wins.

- **Where normalisation starts to pay.** Hertz (a) is the only sample at an intermediate magnitude ratio (3.93), and normalisation is still harmful there ( $\kappa_2$ :  $1.8 \times 10^3 \rightarrow 1.6 \times 10^5$ ). The ratio at which the per-snapshot criterion begins to earn its keep is thus bounded well above 4 — the physics set (604) remains the only evidence that it ever does.
- **Train-only metrics hide the main effect.** With held-out parameters, CPG/ADG’s test error is flat from  $R = 12$  to  $R = 38$  on Hertz: 26 generators improve training error  $4\times$  and generalisation not at all. Only MCPG keeps improving out of sample. Any future knob should be judged against held-out error at matched  $R$ , not training error at its own stopping point.

**Reproducibility.** The two sphere schematics (Figs. 1–2) are illustrative renderings of the normalisation geometry, not data plots: `docs/figures/adg_sphere_2d.png` and `adg_sphere_3d.png`. Matched- $R$  comparison (Table 3): `results/physics/reduction/physics_component_sweep_metrics.csv` and `results/lambda/component_sweep/component_sweep_metrics.csv`, from `python -m greedy.pipelines.pl` and `python -m greedy.pipelines.component_sweep`. Stopping-point runs: `results/{lambda,physics}/reduc` and `results/lambda/adg/`. Normalisation study (Fig. 3): `results/physics/adg_normalization/` and `results/lambda/adg_normalization/`. All numbers are taken verbatim from the `*.csv` and report files there.

Two caveats a reader re-running this should know. (i) The script that produced the two `adg_normalization/` directories was never committed, so they cannot be regenerated from the repository; Fig. 3 rests on them. The effect they illustrate is independently reproducible from `physics_component_sweep_metrics.csv`, which is generated by a committed pipeline and carries the same conclusion at matched  $R$ . (ii) `results/synthetic/` was, until this revision, irreproducible: the snapshot generator drew from the unseeded global `np.random`, so every run produced different data while `pipelines.synthetic’s rng_seed` only fixed the train/test split. The generator now takes an explicit seed and the directory is deterministic; its numbers therefore differ from any pre-revision copy, and no claim in this note depends on them.

Contact-force sets (Sec. 6): `results/contact_forces/` (`comparison_metrics.csv`, `adg_variants.csv`, `matched_r.csv`, `matched_r.png`, `convergence_vs_pod.png`), regenerated with

```
python -m greedy.pipelines.contact_forces_compare --case hertz \
    --epsilon 0.01 --adg-study --matched-r
python -m greedy.pipelines.contact_forces_compare --case membrane \
    --epsilon 0.25 --adg-study --matched-r
```

Hertz cases (a) and (b) (Sec. 6.5): `results/contact_forces/hertz_{a,b}_eps{0.05,0.01,0.001}/`, regenerated with

```
for case in hertz_a hertz_b; do
  for eps in 0.05 0.01 0.001; do
    python -m greedy.pipelines.contact_forces_compare --case $case \
      --epsilon $eps --adg-study --matched-r \
      --output-dir results/contact_forces/${case}_eps${eps}
  done
done
```

Dataset provenance and the  $W$ -norm integration are documented in `docs/contact_force_datasets.md`; the correctness properties the transform relies on are pinned in `tests/test_contact_forces.py`, including the  $P_1$  arc-mass reconstruction of  $W$  that cases (a) and (b) depend on.

**A note on the timing columns.** The `elapsed_seconds` column of `comparison_metrics.csv` was long unusable: the three methods are fitted in a fixed order, and the first one run absorbed the cost of spawning a `joblib` worker pool ( $\approx 1.6$  s), which made CPG look uniformly slower than it is. The parallelism turned out to be a pessimisation on every workload here — each per-row NNLS solve takes microseconds while `joblib` pickles the basis matrix per row — so it was removed; serial is 5–260 $\times$  faster with bit-identical results. The column is now a fair comparison, and it says something the old numbers hid: MCPG is the *slowest* of the three (membrane,  $\varepsilon = 0.25$ : CPG 0.68 s, ADG 0.18 s, MCPG 5.51 s), because its constrained cone-shift QP runs once per generator. That cost is the price of the conditioning win in Table 4.